

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT1-Assignment

View Only

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling
10 marks

Q2

Hospital Patient Registration –
User-Defined Exception
10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

Q5

Manufacturing Plant –
Producer-Consumer Problem
10 marks

5/5 answered

Q2

Hospital Patient Registration – User-Defined Exception

10
marks

Develop a Java program for a hospital patient registration system. The program accepts the patient's name, age, and body temperature. Create suitable user-defined exception classes for invalid age and invalid temperature. Validate the input before registration. Use try-catch-finally and display a registration completion message from finally.

Sample Test Cases:

Input / Scenario Expected Result Purpose

Arun, 35, 98.6 Patient registered successfully Normal case

Meena, 12, 98.4 InvalidAgeException Boundary / exception case

Ravi, 45, 105.2 InvalidTemperatureException Invalid input

Kumar, -2, 98.1 InvalidAgeException Invalid input

Your Code

```
1 import java.util.*;  
2 class Hospital{  
3     public static void main(String[] args){  
4         Scanner sc = new Scanner(System.in);  
5         try{
```

```
6         String name=sc.next();
7         int age=sc.nextInt();
8         double temp=sc.nextDouble();
9         if(age>18)
10            throw new InvalidAgeExcept
11            if(temp<95||temp>105)
12                throw new InvalidTemperatu
13            System.out.println("Patient Re
14        }catch(InvalidAgeException e){
15            System.out.println("InvalidAge
16        }catch(InvalidTemperatureException
17            System.out.println("InvalidTer
18        }finally{
19            System.out.println("Registrati
20        }
21    }
22 }
23 class InvalidAgeException extends Exceptio
24 }
25 class InvalidTemperatureException extends
26 }
```

Input

Arun 35 98.6

Output

InvalidAgeException
Registration Completed

✓ Submitted